CHAPTER 5

The Hartree–Fock Method

5.1 Introduction

The success of the phenomenologically introduced shell model justifies the assumption that nucleons move independently in an average potential produced by all of the nucleons. The question now is how to extract such a single-particle potential out of the sum of two-body interactions

$$V(1\ldots A)=\sum_{i<j=1}^{A} V(i,j)\simeq\sum_{i=1}^{A} V(i), \tag{5.1}$$

and how well this single-particle potential will agree with those used up to now, for example, the harmonic oscillator, the square well, or the Woods–Saxon potential. It will be shown that we can derive a single-particle potential from the two-body interaction by a variational principle using Slater determinants as trial wave functions.

In Section 5.2 we will discuss in general the variational method, which will be important in many of the following chapters. In Section 5.3, we describe the Hartree–Fock method in detail and in Section 5.4 we give an application to a very simple model. In Section 5.5 we treat symmetries in connection with the Hartree–Fock method, and in Section 5.6 we present the Hartree–Fock theory with density dependent forces.

5.2 The General Variational Principle

We first want to show that the exact Schrödinger equation

$$H|\Psi\rangle = E|\Psi\rangle \tag{5.2}$$

is equivalent to the variational equation

$$\delta E[\Psi] = 0, \tag{5.3}$$

with

$$E[\Psi] = \frac{\langle \Psi | H | \Psi \rangle}{\langle \Psi | \Psi \rangle}. \tag{5.4}$$

The variation (5.3) can be obtained from (5.4):

$$\langle \delta\Psi | H - E | \Psi \rangle + \langle \Psi | H - E | \delta\Psi \rangle = 0. \tag{5.5}$$

Since $|\Psi\rangle$ is, in general, a complex function, we can carry out the variation over the real and imaginary part independently, which is equivalent to carrying out the variation over $|\delta\Psi\rangle$ and $\langle\delta\Psi|$ independently. To see this we use the fact that Eq. (5.5) is valid for arbitrary infinitesimal $|\delta\Psi\rangle$. We can replace $|\delta\Psi\rangle$ by $i|\delta\Psi\rangle$ and get

$$-i\langle \delta\Psi | H - E | \Psi \rangle + i\langle \Psi | H - E | \delta\Psi \rangle. \tag{5.6}$$

Together with Eq. (5.5), we find

$$\langle \delta\Psi | H - E | \Psi \rangle = 0 \tag{5.7}$$

and the complex conjugate equation. Since $|\delta\Psi\rangle$ is arbitrary, Eq. (5.7) is equivalent to the eigenvalue problem (5.2).

The approximation of such variational methods consists of the fact that $|\Psi\rangle$ is usually restricted to a set of mathematically simple trial wave functions. As soon as the true function is not in this set, the minimal solution is no longer the exact eigenfunction, but only an approximation. The variational method is especially well suited for determining the ground state, since for any trial wave function $|\Psi\rangle$ we can show that

$$E[\Psi] \geqslant E_0, \tag{5.8}$$

and thus E_0 will always be the lower bound of a variational calculation. To prove this, we develop the trial wave function in terms of the exact eigenfunctions of the Hamiltonian:

$$|\Psi\rangle = \sum_{n=0}^{\infty} a_n |\Psi_n\rangle \tag{5.9}$$

with

$$H|\Psi_n\rangle = E_n|\Psi_n\rangle. \tag{5.10}$$

This yields

$$E[\Psi]=\frac{\sum_{nn'} a_n^* a_n E_n \delta_{nn'}}{\sum_n |a_n|^2} \geqslant \frac{\sum_n |a_n|^2 E_0}{\sum_n |a_n|^2} = E_0, \tag{5.11}$$

which is precisely Eq. (5.8). In cases where the ground state energy is not degenerate, the equality sign in (5.11) is valid, if and only if all the coefficients a_n with $n \neq 0$ vanish, that is, $|\Psi\rangle$ is proportional to $|\Psi_0\rangle$. If we are interested in the first excited state, we then have to carry out the variation within the subspace entirely orthogonal to $|\Psi_0\rangle$, that is, over all the wave functions $|\Psi\rangle$ with $a_0=0$. Within this subspace $|\Psi_1\rangle$ has the minimal expectation value of H. To find $|\Psi_1\rangle$, we must carry out the variation with the subsidiary condition $\langle\Psi_1|\Psi_0\rangle=0$. In principle we can continue and calculate the whole spectrum using this method.

In practice, however, we do not know $|\Psi_0\rangle$ exactly. From a variation in a restricted subset of the Hilbert space, we find only an approximation $|\Phi_0\rangle$. For the calculation of an approximation $|\Phi_1\rangle$ to the first excited state $|\Psi_1\rangle$, we have to solve the variational equation (5.3) with the supplementary condition that $|\Phi_1\rangle$ is orthogonal to $|\Phi_0\rangle$:

$$\langle\Phi_1|\Phi_0\rangle=0. \tag{5.12}$$

For the second excited state, we must have two supplementary conditions, namely:

$$\langle\Phi_2|\Phi_1\rangle=0; \quad \text{and} \quad \langle\Phi_2|\Phi_0\rangle=0. \tag{5.13}$$

These supplementary conditions are coupled to the problem via Lagrange parameters. We thus see that for higher excited states this method quickly gets rather complicated, therefore it has been applied mainly for the calculation of the ground state. Sometimes, however, these conditions are simply fulfilled because of symmetry properties, as, for example, is the case for states with different angular momentum quantum numbers. We will see in Chapter 7 how to calculate a whole rotational band where the determination of each level is no more complicated than that of the ground state.

So far we have shown that for a certain trial wave function, the ground state energy is always larger than or equal to the exact ground state energy and corresponds to an extremum. In actual calculations, we have to make sure that this extremum actually corresponds to a *minimum*, that is, we must calculate the second derivative of the energy functional, for example, with respect to certain parameters. In the case of the Hartree–Fock or Hartree–Fock–Bogoliubov theory, we will come back to this point (in Chapter 7).

In order to decide which of two variational approaches (i.e., two sets of trial wave functions) is the better one, we have two criteria:

(i) If one set of the trial wave functions is a subset of the other, the larger set is usually the better one, because it contains the first's set minimum.

(ii) Since the exact E_0 is a lower bound, we may hope that out of two trial wave functions, the one for which the corresponding energy is closest to E_0 is better.

Both criteria are, however, not exact statements. Pathological examples can be found which contradict them.

We finish this section with the remark that the variation principle is only valid in this form for a linear eigenvalue problem of the type (5.2). In cases where the Hamiltonian itself depends on the wave function we want to determine, we have to be very careful in applying this principle.

5.3 The Derivation of the Hartree–Fock Equation

5.3.1 The Choice of the Set of Trial Wave Functions

Using the fact that the shell model has provided a suitable approximation for the qualitative explanation of many nuclear properties, we shall assume that there is an average single-particle potential (later to be called the *Hartree–Fock potential*)

$$H^{\mathrm{HF}} = \sum_{i=1}^{A} h(i) \tag{5.14}$$

whose eigenfunction having the lowest eigenvalue E_0^{HF} is an approximation to the exact ground state function. This eigenfunction $\Phi(1 \ldots A)$ is, as we have seen in Chapter 2, a *Slater determinant*

$$|\mathrm{HF}\rangle = |\Phi(1 \ldots A)\rangle = \prod_{i=1}^{A} a_i^+ |-\rangle \tag{5.15}$$

in which the Fermion operators a_k^+, a_k correspond to the single-particle wave functions φ_k, which are themselves eigenfunctions of the single-particle Hamiltonian h, viz:

$$h(i)\varphi_k(i) = \epsilon_k \varphi_k(i), \qquad i = \{\mathbf{r}_i, s_i, t_i\}. \tag{5.16}$$

As we have seen in Section 2.5, we obtain the lowest eigenvalue of H^{HF} if one occupies the A lowest levels in the state $|\mathrm{HF}\rangle$ (Eq. 5.15). In the following, we will characterize the occupied levels in $|\mathrm{HF}\rangle$ by the letters i, j (hole states) and the empty levels by m, n (particle states). If we do not distinguish, we use the letters k, l, p, q.

The wave functions $\varphi_k(\mathbf{r}, s, t)$ are a coordinate space representation of the eigenstates $|k\rangle$ of the single-particle Hamiltonian h. Very often, we work in a configuration space based on some arbitrary complete and orthogonal set of single-particle wave functions $\{\chi_l\}$ (an example is the set of spherical harmonic oscillator wave functions). The function φ_k can be

expanded on this basis:

$$\varphi_k = \sum_l D_{lk} \chi_l. \tag{5.17}$$

If, for each wave function χ_l, we define corresponding fermion creation and annihilation operators c_l^+, c_l (see Section C.1), we can similarly express the operators a_k^+ by the operators c_l^+:

$$a_k^+ = \sum_l D_{lk} c_l^+ . \tag{5.18}$$

Since both sets $\{\varphi_k\}$ and $\{\chi_l\}$ are complete and orthogonal, the transformation D has to be unitary:

$$D^+ D = D D^+ = 1. \tag{5.19}$$

This fact also guarantees that the operators (a_k^+, a_k) and (c_l^+, c_l) both obey separate Fermi commutation relations.

As discussed in Section D.2, there is no one-to-one correspondence between a Slater determinant Φ of the form (5.15) and the set of single-particle states φ_k. Any unitary transformation which does not mix particle and hole states leaves Φ unchanged (at least up to an unimportant phase).

It is therefore more convenient to represent a Slater determinant $|\Phi\rangle$ by its single-particle density matrix (D.9):

$$\rho_{ll'} = \langle \Phi | c_{l'}^+ c_l | \Phi \rangle. \tag{5.20}$$

From Eqs. (5.18) and (5.19), we get

$$\rho_{ll'} = \sum_{kk'} D_{lk} D_{l'k'}^* \langle \Phi | a_{k'}^+ a_k | \Phi \rangle = \sum_{i=1}^{A} D_{li} D_{l'i}^* \tag{5.21}$$

because ρ is diagonal in the basis a_k^+, a_k with the eigenvalues (occupation numbers) 1 for $i \leqslant A$ (holes) and 0 for $i > A$ (particles). The trace of ρ is equal to the particle number.

As we show in Appendix D.2, there is a one-to-one correspondence between the Slater determinant Φ and its single-particle density ρ. Single-particle densities ρ of Slater determinants are characterized by the fact that they have only eigenvalues 0 or 1, that is,

$$\rho^2 = \rho. \tag{5.22}$$

ρ is therefore a projector in the space of single-particle wave functions onto the subspace spanned by the hole states φ_i.

In the same way, we can define a projector σ

$$\sigma = 1 - \rho \tag{5.23}$$

onto the subspace spanned by the particle states φ_m.

The *Hartree-Fock method* [Ha 28, Fo 30] is now defined in the following way. We use the set of Slater determinants $\{\Phi\}$ of the form (5.15) consisting of A arbitrary but orthogonal single-particle wave functions φ_i as trial wave functions and minimize the energy within this set. An equivalent statement would be that we use the set of all wave functions

$\{\Phi\}$ whose single particle density (5.20) has the property $\rho^2=\rho$ and $\mathrm{Tr}\,\rho=A$.

As we will see in the following sections, this variation will give us the possibility of determining the single-particle operator H^{HF}.

5.3.2 The Hartree–Fock Energy

Before we are able to carry out the variation which allows us to determine the HF-wave function Φ, we have to calculate the HF-energy

$$E^{\mathrm{HF}}=\langle\Phi|H|\Phi\rangle. \tag{5.24}$$

We start with the many-body Hamiltonian H and represent it in second quantization by the basis operators c_l^+, c_l (see Section C.1):

$$H=\sum_{l_1 l_2} t_{l_1 l_2} c_{l_1}^+ c_{l_2}+\frac{1}{4}\sum_{l_1 l_2 l_3 l_4}\bar{v}_{l_1 l_2, l_3 l_4} c_{l_1}^+ c_{l_2}^+ c_{l_4} c_{l_3}, \tag{5.25}$$

where

$$\bar{v}_{l_1 l_2, l_3 l_4}=v_{l_1 l_2 l_3 l_4}-v_{l_1 l_2 l_4 l_3}. \tag{5.26}$$

Wick's theorem (Sec. C.4) allows us to calculate the energy (5.24) as a functional of the single-particle density

$$\begin{aligned} E^{\mathrm{HF}}[\rho] &= \sum_{l_1 l_2} t_{l_1 l_2}\langle\Phi|c_{l_1}^+ c_{l_2}|\Phi\rangle+\frac{1}{4}\sum_{l_1 l_2 l_3 l_4}\bar{v}_{l_1 l_2 l_3 l_4}\langle\Phi|c_{l_1}^+ c_{l_2}^+ c_{l_4} c_{l_3}|\Phi\rangle \\ &= \sum_{l_1 l_2} t_{l_1 l_2}\rho_{l_2 l_1}+\frac{1}{2}\sum_{l_1 l_2 l_3 l_4}\rho_{l_3 l_1}\bar{v}_{l_1 l_2 l_3 l_4}\rho_{l_4 l_2} \qquad (5.27) \\ &= \mathrm{Tr}\,(t\rho)+\frac{1}{2}\mathrm{Tr}_1\mathrm{Tr}_1(\rho\bar{v}\rho), \qquad (5.28) \end{aligned}$$

where $\mathrm{Tr}_1\mathrm{Tr}_1\ldots$ is an obvious shorthand notation. Eq. (5.28) does not depend on the basis. We can therefore use it to give an expression for the HF-energy in the HF-basis $\{\varphi_k\}$ in which ρ is diagonal with the eigenvalues 0 and 1

$$E^{\mathrm{HF}}=\sum_{i=1}^{A} t_{ii}+\frac{1}{2}\sum_{i,j=1}^{A}\bar{v}_{ij,ij}. \tag{5.29}$$

5.3.3 Variation of the Energy

To determine the HF-basis, we have to minimize the energy (5.28) for all product wave functions $|\Phi\rangle$ or for all densities ρ with the property $\rho^2=\rho$. Since a small variation $\rho+\delta\rho$ has to be a projector again, we get

$$(\rho+\delta\rho)^2=\rho+\delta\rho$$

or, up to linear terms in $\delta\rho$,

$$\delta\rho=\rho\,\delta\rho+\delta\rho\,\rho.$$

In the HF-basis, where ρ is diagonal, this means that the particle–particle (pp) and hole–hole (hh) matrix elements of $\delta\rho$ have to vanish, that is,

$$\rho\,\delta\rho\,\rho = \sigma\,\delta\rho\,\sigma = 0. \tag{5.30}$$

To make sure that we stay within the set of Slater determinants, therefore, we can only allow for variations $\delta\rho_{mi}$ and $\delta\rho_{im}$ of the ph and hp matrix elements of ρ in the HF-basis.

The variation of the energy (5.27) is then given by

$$\delta E = E[\rho+\delta\rho] - E[\rho] = \sum_{kk'} h_{kk'}\delta\rho_{k'k} = \sum_{mi} h_{mi}\delta\rho_{im} + c.c., \tag{5.31}$$

where the Hermitian matrix h is defined as

$$h_{kk'} = \frac{\partial E^{\mathrm{HF}}[\rho]}{\partial\rho_{k'k}}. \tag{5.32}$$

From Eq. (5.27), we obtain

$$h = t + \Gamma \tag{5.33}$$

with the *self-consistent field*

$$\Gamma_{kk'} = \sum_{ll'} \bar{v}_{kl'k'l}\rho_{ll'}. \tag{5.34}$$

Since arbitrary values of $\delta\rho_{mi}$ are allowed, we see from Eq. (5.31), that the condition $\delta E = 0$ for the HF-solution means that the ph matrix elements of h have to vanish,

$$h_{mi} = t_{mi} + \sum_{j=1}^{A} \bar{v}_{mjij} = 0 \qquad (\text{for } i \leqslant A, m > A), \tag{5.35}$$

in the basis where ρ is diagonal, that is, h does not mix particle and hole states of ρ and Eq. (5.35) is equivalent to

$$[h,\rho] = [t+\Gamma[\rho],\rho] = 0. \tag{5.36}$$

This is a nonlinear equation, and not easy to solve. It also states that h and ρ can be diagonalized simultaneously. Since the basis in which ρ is diagonal is determined only up to unitary transformations among the occupied levels or among the empty levels, we use this freedom and require that h shall be diagonal. This defines the *Hartree–Fock basis* and converts (5.36) into an eigenvalue problem.

$$h_{kk'} = t_{kk'} + \sum_{i=1}^{A} \bar{v}_{kik'i} = \epsilon_k \delta_{kk'}. \tag{5.37}$$

Considering the fact that this basis is given by the transformation D (5.18), we obtain the set of *Hartree–Fock equations*

$$\sum_{l'} h_{ll'} D_{l'k} = \sum_{l'} \left(t_{ll'} + \sum_{i=1}^{A} \sum_{pp'} \bar{v}_{lp'l'p} D_{pi} D^{*}_{p'i} \right) D_{l'k} = \epsilon_k D_{lk}, \tag{5.38}$$

which represent a Hermitian eigenvalue problem. It is nonlinear because the matrix h depends on the density ρ, that is, on the solution of the

problem. The coefficients D_{lk} found by the solution of these equations determine the single-particle wave functions φ_k [Eq. (5.17)].

We have thus derived a single-particle Hamiltonian

$$H^{\mathrm{HF}} = \sum_{kk'} h_{kk'} a_k^+ a_{k'} = \sum_{kk'} (t+\Gamma)_{kk'} a_k^+ a_{k'}$$
$$= \sum_{kk'} \left(t_{kk'} + \sum_{j=1}^{A} \bar{v}_{kjk'j} \right) a_k^+ a_{k'} = \sum_k \epsilon_k a_k^+ a_k \tag{5.39}$$

with the properties required in Section (5.3.1): The Slater determinant $|\mathrm{HF}\rangle$, where the lowest A levels are occupied, corresponds to an energy E which is stationary against small variations of the wave function.

The single-particle Hamiltonian h contains, besides the kinetic energy t, a self-consistent field Γ (Eq. (5.34)), which depends on the density of the nucleus. It is a one-body field and averages over all two-body interactions. This point will become even clearer in the coordinate representation (Sec. 5.3.4). The energy expectation value of the HF-wave function $|\mathrm{HF}\rangle$ is given by Eqs. (5.29) and (5.37):

$$E_0^{\mathrm{HF}} = \sum_{i=1}^{A} \epsilon_i - \frac{1}{2} \sum_{ij=1}^{A} \bar{v}_{ijij}. \tag{5.40}$$

It is therefore not equal to the sum of single-particle energies [compare the discussion of this point in Sec. (2.8.6)].

5.3.4 The Hartree–Fock Equations in Coordinate Space

To give a better interpretation of the structure of Eq. (5.38), we write it down in the coordinate space. Assuming a local two-body potential which does not depend on spin or isospin, that is, a pure Wigner force (see Sec. 4.2), we find instead of Eq. (5.38):

$$-\frac{\hbar^2}{2m}\Delta\varphi_k(\mathbf{r}) + \sum_{j=1}^{A} \int d\mathbf{r}'\, v(\mathbf{r},\mathbf{r}') \cdot \varphi_j^*(\mathbf{r}') \{ \varphi_j(\mathbf{r}')\varphi_k(\mathbf{r}) - \varphi_j(\mathbf{r})\varphi_k(\mathbf{r}') \}$$
$$= \epsilon_k \varphi_k(\mathbf{r}). \tag{5.41}$$

Defining the local Hartree potential

$$\Gamma_H(\mathbf{r}) = \int d\mathbf{r}'\, v(\mathbf{r},\mathbf{r}') \sum_{j=1}^{A} |\varphi_j(\mathbf{r}')|^2 = \int d\mathbf{r}'\, v(\mathbf{r},\mathbf{r}')\rho(\mathbf{r}') \tag{5.42}$$

and the nonlocal or exchange potential

$$\Gamma_{Ex}(\mathbf{r},\mathbf{r}') = -v(\mathbf{r},\mathbf{r}') \sum_{j=1}^{A} \varphi_j^*(\mathbf{r}')\varphi_j(\mathbf{r}) = -v(\mathbf{r},\mathbf{r}')\rho(\mathbf{r},\mathbf{r}'), \tag{5.43}$$

we find that $\varphi_k(\mathbf{r})$ is the solution of a nonlocal Schrödinger equation

$$\left\{ -\frac{\hbar^2}{2m}\Delta + \Gamma_H(\mathbf{r}) \right\} \varphi_k(\mathbf{r}) + \int d\mathbf{r}'\, \Gamma_{Ex}(\mathbf{r},\mathbf{r}')\varphi_k(\mathbf{r}') = \epsilon_k \varphi_k(\mathbf{r}). \tag{5.44}$$

Equations (5.38) and (5.44) contain a self-consistency problem, since the potentials Γ, Γ_H and Γ_{Ex} depend on the local and nonlocal density $\rho(\mathbf{r})$ and $\rho(\mathbf{r}, \mathbf{r}')$ of the solution. The equations can be solved by iteration,* starting with a set of phenomenological shell model wave functions to calculate Γ_H and Γ_{Ex} as a first step. Another convenient first guess is the Thomas–Fermi expression for the density (see Sec. 13.2.1). From (5.44) we then get new single-particle wave functions, and so on. This procedure is continued until convergence is obtained, that is, the potentials stay constant in two consecutive steps. In this case, Γ is the self-consistent average potential felt by one particle through interactions with all the other particles.

It should be noticed that, starting with a local two-body interaction, the Fock potential Γ_{Ex} (5.43) is nonlocal. This is caused by the Pauli principle and the antisymmetrization of the matrix element (5.26). A variation of simple product wave functions without antisymmetrization yields only the local Hartree potential. Of course, nonlocal two-body interactions give a nonlocal Hartree term, too. On the other hand, if we use a δ-force, then the Fock term is also local† [see Eq. (5.99)].

Since many of the more formal discussions on HF theory will be taken up in Chapter 7, we will not go into more detail here; we wish only to mention that we will treat there the stability of the Hartree–Fock equations, that is, the question of whether the Hartree–Fock solutions correspond to a minimum or a maximum in the energy. We will also present the so-called gradient method for the solution of the HF equations.

In order to familiarize the reader with the concept of the theory presented in this chapter, we will now present a simple model in which all equations can be solved analytically.

5.4 The Hartree–Fock Method in a Simple Solvable Model

As we will discuss in the last section of this chapter, all realistic HF calculations are very difficult numerical problems. In order to get some feeling about how this method works, we want to apply it to a very simple, exactly solvable model first proposed by Lipkin, Meshkov, and Glick [LMG 65], and which has been widely used to test all kinds of many-body theories (as we shall see later on). Let us imagine two levels in a fixed shell model potential having the same j-value, one situated just below the Fermi level, the other just above. The level below the Fermi level is filled with $2j+1$ nucleons (of one kind, for simplicity). The fixed single-particle potential can be thought of as being produced by an especially stable core formed out of the rest of the nucleons. It is then conceivable to calculate solely that part of the average potential coming from the nucleons in the last j-shell, in a self-consistent manner, such that the mutual influence of the core and the last j-shell is neglected. Of course, the idea that there is only one level with the same j above the Fermi level is very unrealistic and serves only to schematize the problem.

* For numerical methods to solve the HF-equations (5.38) or (5.41), see also [QF 78].

† In this case, however, we also have to take into account spin degrees of freedom, otherwise the exchange term cancels the direct term.

Furthermore, it is assumed that in the basis produced by the fixed potential of the core, the residual interaction of the nucleons in the two shells is of a very special form, being of the monopole–monopole type (see Chap. 4) and having essentially only one matrix element different from zero (a particle–hole matrix element of the RPA type (see Chap. 8)). The *model Hamiltonian* is then of the form

$$H=\epsilon K_0-\tfrac{1}{2}V(K_+K_++K_-K_-), \tag{5.45}$$

with

$$K_0=\frac{1}{2}\sum_{m=1}^{\Omega}(c_{+m}^+c_{+m}-c_{-m}^+c_{-m}); \qquad K_+=\sum_{m=1}^{\Omega}c_{+m}^+c_{-m}; \qquad K_-=(K_+)^+, \tag{5.46}$$

where $\Omega=2j+1$ and c_{+m}^+, c_{-m}^+ create a particle in the upper and lower levels, respectively, and ϵ is the energy difference between the two levels (see Fig. 5.1). The operators K_0, $K_\pm$ fulfill the commutation relations of angular momenta.

$$[K_+, K_-]=2K_0; \qquad [K_0, K_\pm]=\pm K_\pm. \tag{5.47}$$

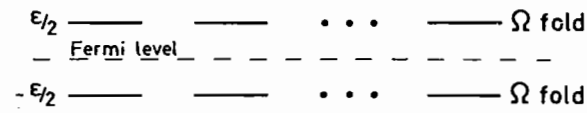


Figure 5.1. Level scheme in the schematic model.

However, it must be emphasized that the operators K_0, $K_\pm$ have nothing to do with rotations in coordinate space. They are often referred to as quasi-spin operators. In order to apply the Hartree–Fock method, we have to construct the general Slater determinant $|\Phi\rangle$. The Hamiltonian (5.45) is invariant under a permutation of the Ω levels below and corresponding levels above the Fermi surface. Therefore, in the following, we restrict ourselves to those solutions of the problem which are completely symmetric under such a permutation. In this case, there is only one possibility of exciting *ph*-pairs, and the most general Slater determinant $|\Phi\rangle$ is characterized by the complex number z:

$$|\Phi\rangle=\mathcal{N}\exp\{zK_+\}|\Phi_0\rangle=\prod_{m=1}^{\Omega}a_{0m}^+|-\rangle, \tag{5.48}$$

with

$$|\Phi_0\rangle=\prod_{m=1}^{\Omega}c_{-m}^+|-\rangle \tag{5.49}$$

and

$$a_{0m}^+=D_{-0}c_{-m}^++D_{+0}c_{+m}^+ \tag{5.50a}$$

$$a_{1m}^+=D_{-1}c_{-m}^++D_{-1}c_{+m}^+, \tag{5.50b}$$

where we denote the new lower and upper levels by 0 and 1. With the aid of relations (5.50a) and (5.50b), we can express the Hamiltonian (5.45) in the new operators a_{0m}^+, a_{1m}^+. Then, varying $\langle\Phi|H|\Phi\rangle$ with respect to D_{-0}^* and D_{+0}^* yields* the Hartree–Fock equations in the Lipkin model (this is, in fact, just another way

* Under the subsidiary condition that the Ds are normalized [see Eq. (5.96)].

to derive the HF eqs.):

$$\begin{pmatrix} -\frac{1}{2} & -Q \\ -Q^* & \frac{1}{2} \end{pmatrix} \begin{pmatrix} D_{-0} \\ D_{+0} \end{pmatrix} = \frac{e_0}{\epsilon} \begin{pmatrix} D_{-0} \\ D_{+0} \end{pmatrix}, \tag{5.51}$$

where

$$\chi = \frac{V}{\epsilon}(\Omega - 1), \qquad Q = \chi D_{+0} D^*_{-0},$$

which is of the usual nonlinear type and can be solved by iteration. The new single-particle energies are given by

$$\begin{aligned} e_{0,1} &= \mp \epsilon \sqrt{\frac{1}{4} + |Q|^2} \\ &= \mp \frac{\epsilon}{2} \sqrt{1 + \chi^2 \sin 2\alpha}\ , \end{aligned} \tag{5.52}$$

where we put $D_{-0} = \cos\alpha$ and $D_{+0} = \sin\alpha \cdot e^{-i\varphi}$. Solving Eq. (5.51) for D_{-0} and D_{+0} and inserting these into the expression for Q yields, with (5.52) in the case $Q \neq 0$, the following equation for the "deformation" potential Q:

$$1 = \frac{\chi}{2} \frac{1}{\sqrt{\frac{1}{4} + Q^2}}\ . \tag{5.53}$$

This equation has only real solutions for $\chi \geqslant 1$ corresponding to the deformed HF solution given below. (The fact that Eq. (5.53) is very similar to the gap equation of BCS theory [Eq. (6.60)] is not accidental, as will be discussed in more detail in Sec. 11.2 and Appendix F.5.)

In the coordinates α, φ, we then obtain for the ground state energy

$$E_0^{\mathrm{HF}} = -\frac{\epsilon}{2}\Omega\left(\cos 2\alpha + \frac{1}{2}\chi \sin^2 2\alpha \cdot \cos 2\varphi\right) \tag{5.54}$$

and the self-consistency condition

$$\begin{aligned} \frac{\partial E_0^{\mathrm{HF}}}{\partial \alpha} &= 0 = \epsilon\Omega \sin 2\alpha (1 - \chi \cos 2\alpha \cdot \cos 2\varphi), \\ \frac{\partial E_0^{\mathrm{HF}}}{\partial \varphi} &= 0 = \epsilon\Omega \frac{1}{2} \sin 2\alpha \cdot \sin 2\varphi. \end{aligned} \tag{5.55}$$

From Eq. (5.55), we see that we have to distinguish two cases, depending on whether χ is greater than or smaller than one. In the latter case, we have only one solution:

$$\varphi_{\mathrm{HF}} = 0, \qquad \alpha_{\mathrm{HF}} = 0, \qquad \chi < 1. \tag{5.56}$$

For $\chi > 1$ we have a second solution,

$$\varphi_{\mathrm{HF}} = 0, \qquad \chi \cdot \cos 2\alpha_{\mathrm{HF}} = 1, \qquad \chi > 1, \tag{5.57}$$

which turns out to correspond to the minimum in the energy, the first solution having been a maximum. This can be seen from the curvature of the energy at this point:

$$\left.\frac{\partial^2 E_0^{\mathrm{HF}}}{\partial \alpha^2}\right|_{\alpha = \alpha_{\mathrm{HF}}} = 2\epsilon\Omega\left(\chi - \frac{1}{\chi}\right) > 0 \qquad \text{for} \quad \chi > 1. \tag{5.58}$$

In Fig. 5.2 we show a cut ($\varphi = 0$) through the two-dimensional energy surface (5.54) for $\chi \lesssim 1$.

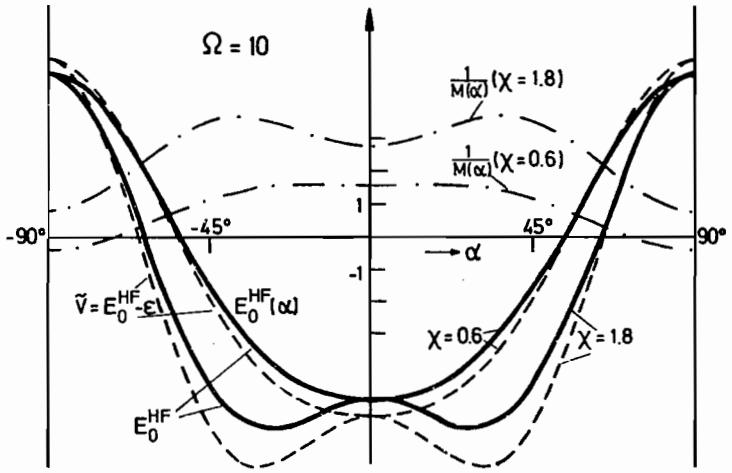


Figure 5.2. E_0^{HF} as a function of α for $\chi \lesseqgtr 1$ (full lines). The other four curves correspond to quantities calculated within the generator coordinate method (see Sec. 10.7.5). (From [Ho 73].)

From the critical value

$$\chi_c = \frac{V_c(\Omega - 1)}{\epsilon} = 1 \tag{5.59}$$

on, the solution at $\alpha = 0$, $\varphi = 0$ becomes unstable and then we have to use a different single-particle basis. In Fig. 5.3, we show for $\chi = 6$ how the iterative solution of equation (5.51) works. Starting with $\alpha_1 = 80°$, $\varphi_1 = 30°$ we find in the subsequent steps (full lines in Fig. 5.3): $\alpha_2 = 32.01°$, $\varphi_2 = -30°$; $\alpha_3 = 39.75°$, $\varphi_3 = 30°$; $\alpha_4 = 40.19°$, $\varphi_4 = -30°$; ... . With respect to α we get a rapid convergence to the solution $\alpha_{HF} = 40.20$ [Eq. (5.57)]; with respect to the variable φ, however, we do not get convergence, the solution jumping back and forth between +30° and

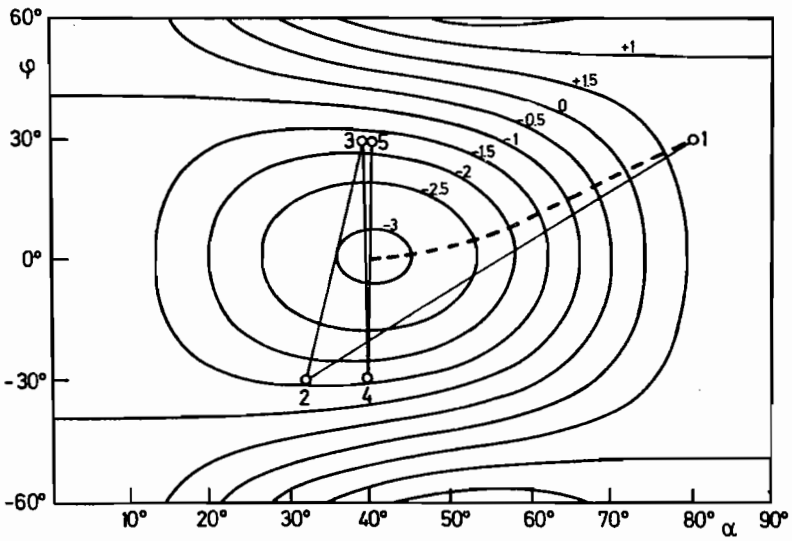


Figure 5.3. Numerical comparison of solution of the HF equations by iterative diagonalization (solid line) and the gradient method (dashed line).

$-30°$. Our model is certainly a very special case in the sense that in the general case the iterative diagonalization converges to a minimum in the energy surface. However, one sometimes also encounters in realistic calculations cases where, within a certain degree of freedom, the solution oscillates back and forth, as in our model. It is therefore preferable to use another method to find the minimum which can, in general, also give faster convergence. This is the gradient method, which will be explained in Section 7.3.3. The way in which the gradient method would converge to the minimum in our model is also indicated in Fig. 5.3 by the dashed line.

5.5 The Hartree–Fock Method and Symmetries

The HF equations (5.38) are nonlinear, which means that the self-consistent potential Γ [Eq. (5.34)] depends not only on the original Hamiltonian, but also on the solution [which is represented by its density matrix ρ (5.21)]. Therefore, this potential does not necessarily show the same symmetries as the Hamiltonian. We then say that the solution has a broken symmetry. The loss of translational invariance is one which we have already encountered, but, as we shall see, there are others—such as, for example, the rotational invariance and particle number conservation (see Chs. 6 and 7). Usually one calls a transition from a symmetry-conserving solution to a symmetry-broken solution a phase transition. The great advantage of this symmetry breaking is the fact that it allows us to take into account, in an approximate way, many-body correlations without losing the simple picture of independent particles. A more detailed discussion of this point will be given in Chapters 7 and 11 and in Appendix F, but let us give here the following argument: Let us suppose that we have a Slater determinant $|\Phi\rangle$ which consists of deformed single-particle wave functions $\{a_k^+\}$. We express them in a spherical basis $\{c_q^+\}$ through the linear transformation

$$a_k^+ = \sum_q D_{qk} c_q^+ . \tag{5.60}$$

We thus obtain a sum of Slater determinants in the spherical basis which differs in the occupation numbers of the level q:

$$a_{i_1}^+ \ldots a_{i_N}^+ |-\rangle = \sum_{q_1 \ldots q_N} d_{q_1 \ldots q_N}^{i_1 \ldots i_N} c_{q_1}^+ \ldots c_{q_N}^+ |-\rangle . \tag{5.61}$$

Here the coefficients $d_{q_1 \ldots q_N}^{i_1 \ldots i_N}$ are the corresponding minors of the matrix D_{qk}. They are of a special form; had we allowed them to be of the most general form, the ansatz (5.61) would contain the exact answer; however, this can never be the case in an independent particle (HF) description. Nevertheless, we see that a Slater determinant in one basis can be a complicated superposition of Slater determinants in another.

Of course, the exact wave function should have the symmetries of the Hamiltonian, and their violation is a matter of convenience in order to maintain the independent particle picture for as long as possible. Neverthe-

less, we should try at a later stage to restore the symmetries. How this can be achieved will be discussed in Chapter 11. However, certain symmetries are conserved in the HF theory, the so-called *self-consistent symmetries*. They depend on the symmetries of the initial density $\rho^{(0)}$ of the iteration, and are based on the property of the HF field Γ viz:

$$S\Gamma[\rho]S^+ = \Gamma[S\rho S^+] \tag{5.62}$$

for any matrix S of a unitary* symmetry operator

$$\hat{S} = \sum_{ll'} S_{ll'} c_l^+ c_{l'} \tag{5.63}$$

which commutes with the many-body Hamiltonian (5.25)

$$[H, \hat{S}] = 0.$$

The property (5.62) is easy to show for the definition (5.34) of Γ:

$$(S\Gamma S^+)_{kk'} = \sum_{ll'qq'pp'} S_{kp} S^*_{k'p'} \langle pq'|V|p'q\rangle_{as} S^*_{lq} S_{l'q'} (S\rho S^+)_{ll'} \tag{5.64}$$

$$= \sum_{ll'} \langle kl'|\hat{S}V\hat{S}^+|p'q\rangle_{as} (S\rho S^+)_{ll'}. \tag{5.65}$$

From the invariance of the two-body interaction

$$\hat{S}V\hat{S}^+ = V \tag{5.66}$$

we finally get (5.62) and

$$Sh[\rho]S^+ = h[S\rho S^+]. \tag{5.67}$$

This means that if the initial density $\rho^{(0)}$ has a certain symmetry S of the Hamiltonian H, then the field $h[\rho^{(0)}]$ for the first step of the iteration has it also. The density $\rho^{(1)}$ is found by a diagonalization of $h[\rho^{(1)}]$. It therefore has to have the same symmetry again, and so on. In each step of the iteration, the symmetry S is conserved.

The existence of these self-consistent symmetries has the following implications for practical calculations [Ri 68].

(i) If we expect a certain symmetry for the solution, we can start with an initial density $\rho^{(0)}$ which has this symmetry and therefore reduce the computational effort by working in a basis consisting of eigenstates of this symmetry.

(ii) If we start with a certain symmetry, we will always stay within this symmetry, and the minimum of energy can only be found among the wave functions that have this symmetry. If, for instance, the deepest minimum is a deformed Slater determinant, we will never get to it starting with a spherically symmetric density matrix, though it may happen that small numerical errors cause small deviations, from the initial symmetry, which grow during the iteration.

*The following considerations also apply, with small modifications, for antilinear unitary symmetries [Me 61].

(iii) If we have a solution with a broken symmetry S

$$S\rho S^+ = \rho_1 \neq \rho, \tag{5.68}$$

then from Eq. (5.67) we obtain the HF-Hamiltonian $h[\rho_1]$, which belongs to the transformed density and has the form

$$h[\rho_1] = S^+ h[\rho] S. \tag{5.69}$$

This shows that ρ_1 is also a solution of the HF-equation (5.36)

$$[h[\rho_1], \rho_1] = 0. \tag{5.70}$$

In the case of a continuous symmetry, like that of translation or rotation, we therefore have, to each symmetry-breaking solution, an infinite number of degenerate solutions.

In the next section, we shall see that it is useful to do HF calculations with density dependent two-body forces. In this case, the force does not necessarily exhibit the same symmetry properties as the bare nucleon–nucleon force. Nevertheless, it is clear that the usual properties of HF solutions with respect to symmetry transformations, as previously discussed, are also retained for density dependent forces, if we require that the interaction satisfies the very plausible condition [BG 77]

$$\hat{S} V[\rho] \hat{S}^{-1} = V[S\rho S^{-1}]. \tag{5.71}$$

This means, for instance, that in the case of rotations the two-body interaction in a rotated system is the same as the interaction calculated with a rotated density. This condition is fulfilled for the Skyrme force (4.109).

5.6 Hartree–Fock with Density Dependent Forces

5.6.1 Approach with Microscopic Effective Interactions

5.6.1.1. Brückner–Hartree–Fock. One of the main obstacles to a direct application of the Hartree–Fock method, outlined in the preceding sections, is the fact that most bare nucleon–nucleon forces have an infinite or at least very repulsive core (see Chap. 4). As is easily verified, the two-body matrix elements entering the Hartree–Fock potential (5.34) all become infinite for a hard core potential. The way to solve this problem is to replace the bare interaction in (5.34) by the Brückner G-matrix discussed in Section 4.3.1. This, as a matter of fact, is not only convenient because it solves the hard core problem, but it can also be shown that it is a consistent resummation of certain higher order terms of the full many body problem. Since we do not have the technical many-body apparatus at hand here, we leave the demonstration to Appendix F.4.

The Brückner Hartree–Fock equations are given in analogy to Eq. (5.38) by (see also the review articles on the subject treated in this section by H.

S. Köhler [Kö 75] and W. Wild [Wi 77]):

$$\sum_{l'}\left\{t_{ll'}+\sum_{i=1}^{A}\sum_{pp'}G_{lp',l'p}^{\epsilon_k+\epsilon_i}D_{pi}D_{p'i}^{*}\right\}D_{l'k}=\epsilon_k D_{lk}. \tag{5.72}$$

Here G is the Brückner G-matrix as defined in Eq. (4.39), written in terms of the basis of a definite single-particle potential. The corresponding ground state energies given in the basis, which are solutions of (5.72), are

$$E_0^{\mathrm{BHF}}=\sum_{i=1}^{A}t_{ii}+\frac{1}{2}\sum_{i,j=1}^{A}G_{ij,ij}^{\epsilon_i+\epsilon_j} \tag{5.73}$$

$$=\sum_{i=1}^{A}\epsilon_i-\frac{1}{2}\sum_{i,j=1}^{A}G_{ij,ij}^{\epsilon_i+\epsilon_j}. \tag{5.74}$$

For the ground state energy, it seems that we need only the hole solutions of (5.72). The particle solutions of (5.72), however, enter the Bethe–Goldstone equation (4.39) through the intermediate particle energies ϵ_n, ϵ_m. The Brückner Hartree–Fock solution then consists of a complicated doubly self-consistent procedure. It can be solved, for example, by the following iteration cycle: (i) Calculate the G-matrix via (4.39) in a suitable basis of first choice (e.g., harmonic oscillator); (ii) diagonalize once (5.72) in this basis, which gives a new basis; (iii) calculate in this new basis a new G-matrix; and so on until the convergence is achieved. In this iteration cycle there arises, however, a small ambiguity concerning the energy dependence of the G-matrix in (5.72). Since we do not know the solution a priori, we have to include the energy dependence of $G_{lp',l'p}^{\epsilon_k+\epsilon_i}$ into the iteration cycle. We thus have to take for ϵ_k the energy corresponding to the basis in which we have actually written Eq. (5.72) for each step of the iteration, that is, we can take ϵ_k equal to ϵ_l or $\epsilon_{l'}$. No ambiguity arises for ϵ_i because the D_{pi} are taken to be diagonal in the iteration process. The conventional choice for the BHF potential energy matrix [see Eq. (5.34)] is (see e.g. [Ba 69a]) (of course, the final answer does not depend on any specific convention):

$$\Gamma_{kk'}^{\mathrm{BHF}}=\begin{cases}\dfrac{1}{2}\displaystyle\sum_{i=1}^{A}\left(G_{ki,k'i}^{\epsilon_k+\epsilon_i}+G_{ki,k'i}^{\epsilon_{k'}+\epsilon_i}\right) & \text{for} \quad k,k'\leqslant\epsilon_F,\\ \displaystyle\sum_{i=1}^{A}G_{ki,k'i}^{\epsilon_k+\epsilon_i} & \text{for} \quad \begin{matrix}k\leqslant\epsilon_F\\ k'>\epsilon_F\end{matrix},\\ \displaystyle\sum_{i=1}^{A}G_{ki,k'i}^{\epsilon_{k'}+\epsilon_i} & \text{for} \quad \begin{matrix}k'\leqslant\epsilon_F\\ k>\epsilon_F\end{matrix}.\end{cases} \tag{5.75}$$

The particle–particle matrix elements of Γ^{BHF} are a somewhat controversial matter. For a fixed G-matrix, the particle–particle matrix elements of Γ^{BHF} do not influence the hole solutions of Eq. (5.72); they do influence, however, the particle solutions, and via (4.39) in the doubly self-consistent cycle, indirectly also the hole solutions. We can argue (see discussions in

[Ba 69a, Ne 70]) that if the particle–particle matrix elements of Γ^{BHF} are equal to zero,

$$\Gamma^{BHF}_{kk'}=0, \qquad \text{for} \quad k,k'>\epsilon_F, \tag{5.76}$$

three-body correlations are effectively summed, which were originally not present in our formulation (5.72) and (4.39).

Equation (5.72) is not only different from the ordinary HF equation (5.38) because the two body operator is more complicated, but also because Γ^{BHF} now depends on the energy which we want to calculate. Therefore, it is a nonlinear problem in which the solutions to different energies are, in general, no longer orthogonal (solutions with different angular momentum are, though, still orthogonal for the spherically symmetric case). However, for the iterative solution this is of no special importance.*

As we discussed in Section 4.3.1, the G-matrix sums up two-particle scattering processes in the nuclear medium. One can show that (5.74) contains all contributions of this type to the ground state energy [Ma 67b]. In Table 5.1 we show the results of Brückner–Hartree–Fock calculations for the rms radius and the binding energy per particle in the case of $^{16}_{8}O$, $^{40}_{20}Ca$, and $^{208}_{82}Pb$; the bare force was the Reid soft core potential (4.35).

Table 5.1 Results of BHF calculations with the Reid soft core potential compared with experiment (from [DMS 73])

		BHF	Experiment
$^{16}_{8}O$	$-E_0/A$ (MeV)	3.91	7.98
	rms (fm)	2.50	2.73
$^{40}_{20}Ca$	$-E_0/A$	3.88	8.55
	rms	3.04	3.48
$^{208}_{82}Pb$	$-E_0/A$	2.52	7.87
	rms	4.51	5.50

The results in Table 5.1 are deceiving. The calculations do not give even half the experimental binding energy and the rms radii are about 10–20% off. The bad result for the binding energy is not too surprising, however, in view of the fact that it is a difference between two very large numbers, that for kinetic and that for potential energy.

Nevertheless, in view of the unsatisfying result, one has to envisage taking into account more complicated processes.† Next in the hierarchy are three-particle scattering terms whose importance should depend on the density of the system. As we have seen in Section 4.3, the healing distance, which characterizes the range of the two-particle correlations, is appreciably smaller than the average interparticle distance. Thus the probability

* One should nevertheless make sure that the energy dependence of Γ^{BHF} is not too strong, otherwise the independent particle picture may no longer be valid.

† In this context, see also the hypernetted chain formalism [PB 73, FR 75, 76a, LS 77, Ri 79].

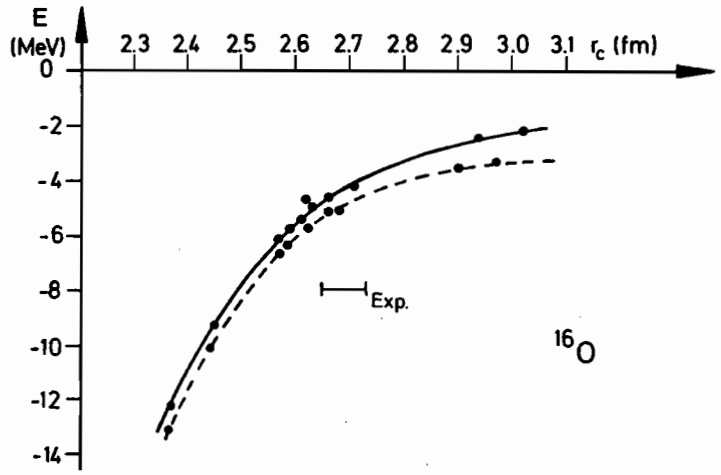


Figure 5.4. Comparison of BHF calculations with different nucleon–nucleon potentials (dots on full line) with calculations including 3-body correlations (broken line) and with experiment (from [Wi 77]).

that three particles are simultaneously within the range of the healing distance should be quite small. This is, however, a rather qualitative argument [Be 71, p. 161]. In fact, Kümmel and co-workers have been able to include three-body correlations in a completely self-consistent manner into the calculations for the gound state energy (see, e.g., [KLZ 75, 78] and further references cited therein). They also give strong arguments that four-body correlations should be negligible. Their results are very interesting in the sense that agreement with experiment is improved, though not very dramatically. For different nucleon–nucleon potentials, the results lie roughly on a smooth line in the $-E_0/A$ versus rms-radius plane, as shown in Fig. 5.4. This line is called the *Coester line*. The results of the Reid soft core potential are located approximately at the point where the curve is closest to the experimental value. In view of the claim [KLZ 75] that the results have converged, the remaining difference must be attributed to relativistic effects or to mesonic degrees of freedom.

5.6.1.2. The Local Density Approximation and the Variational Principle. In Section 4.3 we saw that the influence of a hard core potential on a two-nucleon wave function is effective over relative distances of the two nucleons ranging from 0 to about 1 fm, that is, only over very short distances, compared to the nuclear radius of about 10 fm. This means that the Brückner G-matrix is only different from the bare interaction $v(r)$ within the same range of r-values. However, in general, variations of about 1 fm in the relative variable $r=|\mathbf{r}_1-\mathbf{r}_2|$ of the two nucleons imply a similar variation in the center of mass coordinate. The nuclear density $\rho(\mathbf{R})=\rho(\frac{1}{2}(\mathbf{r}_1+\mathbf{r}_2))$ does not (at least, not in the interior of the nucleus) change very much over such a range of $\mathbf{R}$-values. It therefore seems a reasonable assumption to calculate the nuclear G-matrix at each value of $\rho(\mathbf{R})$ as if the nucleus locally around $\mathbf{R}$ were a piece of nuclear matter* with the density $\rho(\mathbf{R})$. Actually, this reasoning is

* For the theory of nuclear matter, see the reviews [Be 71, Sp 72, JLM 76].

very close to the Thomas–Fermi approximation (see Chap. 13). More explicitly, this means that in the Bethe–Goldstone equation (4.39), which can be written using (5.76) as

$$G^W = \bar{v} + \bar{v}\frac{Q_F}{W - Q_F \Gamma Q_F} G^W, \tag{5.77}$$

we replace the projector Q_F by its expression Q_F^{NM} in nuclear matter of density $\rho(\mathbf{R})$. Since Q_F (4.41) can be expressed by the nonlocal single-particle density $\rho(\mathbf{r}, \mathbf{r}') = \sum_{i=1}^{A} \varphi_i(\mathbf{r})\varphi_i^*(\mathbf{r}')$,

$$\langle \mathbf{r}_1\mathbf{r}_2 | Q_F | \mathbf{r}_1'\mathbf{r}_2' \rangle = [\delta(\mathbf{r}_1 - \mathbf{r}_1') - \rho(\mathbf{r}_1, \mathbf{r}_1')][\delta(\mathbf{r}_2 - \mathbf{r}_2') - \rho(\mathbf{r}_2', \mathbf{r}_2')], \tag{5.78}$$

this means that in (5.78) ρ has to be replaced by its nuclear matter value [see Eq. (13.20)]

$$\rho^{\mathrm{NM}}(\mathbf{r}, \mathbf{r}') = \frac{3}{k_F(\mathbf{R})s} j_1(k_F(\mathbf{R})s)\rho(\mathbf{R}), \tag{5.79}$$

where $\mathbf{s} = \mathbf{r} - \mathbf{r}'$, $\mathbf{R} = \frac{1}{2}(\mathbf{r} + \mathbf{r}')$, and j_1 is a Bessel function. The **R**-dependence of (5.79) comes from the fact that we have only treated the nucleus locally around **R** as nuclear matter. The relationship between k_F and ρ is given, as usual, by [see Eqs. (5.91), (13.22), and (13.23)]

$$\rho(\mathbf{R}) = \frac{2}{3\pi^2} k_F^3(\mathbf{R}). \tag{5.80}$$

Numerical comparison with exact density matrices of finite nuclei have shown [Ne 75] that the nonlocal behavior of ρ is very well approximated by (5.79) in the nuclear interior, and reasonably well represented throughout the nuclear surface. We can therefore conclude that the local density approximation to the G-matrix as described here is quite good.

In principle, the G-matrix thus calculated depends on the three variables $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$; $\mathbf{r}' = \mathbf{r}_1' - \mathbf{r}_2'$ and on $\mathbf{R} = \frac{1}{2}(\mathbf{r}_1 + \mathbf{r}_2 - \mathbf{r}_1' - \mathbf{r}_2')$; $G = G_{\mathbf{r}, \mathbf{r}'; \mathbf{R}}$. It turns out, however, that the dependence of G on the total momentum **P** (the conjugate variable to **R**) is weak, and we usually put it equal to some average value [BGW 58]:

$$G_{\mathbf{r}, \mathbf{r}', \mathbf{P}} \simeq G_{\mathbf{r}, \mathbf{r}', \mathbf{P}_{av}}.$$

Negele [Ne 70] further simplifies the expression for G. He replaces the starting energy $W = \epsilon_i + \epsilon_j$ appearing in the G-matrix of Eq. (5.73) by an average value obtained as twice the average hole energy in nuclear matter. In this way, we get rid of the state dependence of the single-particle potential in Eq. (5.72) and the corresponding eigenfunctions form a complete orthonormal set. The next step consists of representing $G_{\mathbf{r}, \mathbf{r}'}$ by an effective local operator $v^{\mathrm{eff}}(r)$; (we omit the details of how this is achieved and refer to [Ne 70]*). Furthermore, this effective interaction is renormalized phenomenologically to give the correct volume of nuclear matter [see the Bethe–Weizsäcker formula (1.4)]. In this sense, the "density dependent HF" (DDHF) method of Negele is still a semi-phenomenological procedure.

One further very important ingredient of Negele's theory we have not spoken of until now: In Table 5.1, we have seen that a pure BHF calculation does not give very good results; therefore, something better has to be invented. It turns out that

* Other applications of the 'local' density approximation can be found in [Kö 65, NV 70, CS 72, NR 72].

the intuitively appealing idea to vary the ground state energy (5.73) with G replaced by the local density approximation, with respect to the single-particle wave functions, as in the pure HF case (Eq. 5.29), yields very good results. The Hartree–Fock Hamiltonian becomes, in this case, [see Eq. (5.32)]

$$h_{kk'}^{\mathrm{HF}} = \frac{\delta E[\rho]}{\delta \rho_{k'k}} = t_{kk'} + \sum_{qq'} \bar{v}_{kq'k'q}^{\mathrm{eff}} \rho_{qq'} + \langle \Phi | \frac{\delta v^{\mathrm{eff}}}{\delta \rho_{k'k}} | \Phi \rangle, \tag{5.81}$$

where $|\Phi\rangle$ is a Slater determinant corresponding to the density ρ. For density independent forces, (5.81) just reduces to the usual HF-Hamiltonian. For density dependent forces, however, we have an additional term $\delta v^{\mathrm{eff}}/\delta\rho$, usually called the *rearrangement* or *saturation potential*. It results from the density dependence in $v^{\mathrm{eff}}(\rho)$ and plays a very important role in practical calculations. It has to be emphasized, however, that the variational method just described is not equivalent to the variation of a trial wave function minimizing the expectation value of the original many-body Hamiltonian (5.25). This implies that, at least in principle, the ground state energy calculated with (5.81) could be lower than the exact one. The method is justified in the first place by its success; theoretically, we can say that the additional term $\delta v^{\mathrm{eff}}/\delta\rho$ somehow takes into account three-body scattering terms [Ne 75] ($\delta v^{\mathrm{eff}}/\delta\rho$ has six indices in the shell model space). In Table 5.2, we show the big improvement over BHF which is obtained when using the DDHF method [Ne 70, NV 75].

Table 5.2 Binding energies (in MeV) and rms radii (in fm) with the density dependent Hartree–Fock method (DDHF)

		BHF	DDHF	Exp.
$^{16}_{8}$O	$-E_0/A$	3.91	7.59	7.98
	rms	2.50	2.75	2.73
$^{40}_{20}$Ca	$-E_0/A$	3.88	7.99	8.55
	rms	3.04	3.46	3.48
$^{208}_{82}$Pb	$-E_0/A$	2.52	7.83	7.87
	rms	4.51	5.49	5.50

The solution of the DDHF equations is a very complicated task because of the fact that the exchange term (5.43) implies that an integrodifferential equation must be solved. The fact that the nuclear force is of rather short range can, however, be exploited to simplify the problem so that the nonlocal exchange term can be expanded in powers of the nonlocality around its local value. This procedure leads in a rather natural way to a justification of the very successful HF scheme using the phenomenological Skyrme forces, which we will study next [Ne 75].

5.6.2 Hartree–Fock Calculations with the Skyrme Force

5.6.2.1. General Remarks. There have been many applications of the Hartree–Fock method over the years using different forces. It lies beyond the scope of this book to give a review of them (for a recent review, see [QF 78]). Many of them—in particular, those that do not use density

dependent forces—have the shortcoming that they are not able to simultaneously produce the binding energies, the radii, and the proper single-particle spectrum for light and heavy nuclei (see, for instance, [Vo 65, MB 65, BB 67, NDK 68, PS 68, SFW 69, SP 70]).

The use of the Skyrme force (4.104) as a phenomenological interaction in HF calculations has the important merit of being able to very well reproduce binding energies and nuclear radii over the entire periodic table.* Like the Negele force, obtained from the local density approximation, the Skyrme force is density dependent and, as discussed in the last section, cannot simply be used in the variation principle (5.3), which is based on a linear Hamiltonian. Formally, this difficulty can be overcome by considering the Skyrme force as a three-particle force (4.104).

5.6.2.2. The Energy Calculated with the Skyrme Force. We start with the Skyrme force as given in Eq. (4.104) and calculate the expectation value of the corresponding Hamiltonian with respect to a Slater determinant $|\Phi\rangle$ containing the single-particle wave functions $\varphi_i(\mathbf{r}, s, t)$. For the sake of simplicity we require time reversal invariance. This is not essential [Pa 76], but simplifies the equations considerably. Furthermore, we regard only nuclei with $N=Z$ and neglect the Coulomb field.† (For the case $N \neq Z$ with Coulomb interaction, and for more details of the following derivation, see the paper of Vautherin and Brink [VB 72]).

The energy is given by:

$$E_0 = \langle \Phi | T + V^{(2)} + V^{(3)} | \Phi \rangle$$
$$= \sum_{i=1}^{A} \langle i | \frac{p_i^2}{2m} | i \rangle + \frac{1}{2} \sum_{i,j=1}^{A} \langle ij | \bar{v}^{(2)} | ij \rangle + \frac{1}{6} \sum_{ijk=1}^{A} \langle ijk | \bar{v}^{(3)} | ijk \rangle. \quad (5.82)$$

Because of the δ-function character of the Skyrme force (4.104), it is possible to express E by an integral over an energy density $H(\mathbf{r})$

$$E_0 = \int H(\mathbf{r})\, d^3r, \quad (5.83)$$

in which $H(\mathbf{r})$ is an algebraic function of three quantities:

(i) the nucleon density

$$\rho(\mathbf{r}) = \sum_{i,s,t} |\varphi_i(\mathbf{r}, s, t)|^2; \quad (5.84)$$

(ii) the kinetic energy density

$$\tau(\mathbf{r}) = \sum_{i,s,t} |\nabla \varphi_i(\mathbf{r}, s, t)|^2; \quad (5.85)$$

* There is a large number of similar phenomenological density dependent forces with zero range [Mo 70, LV 71, La 72, EM 72, BJS 75, Kö 76, LB 76, TK 76, KKS 77, SM 78] or finite range [Kr 70, ZR 71, RPS 72, LMV 73, Go 75b, RBP 77] which show properties similar to the Skyrme force in HF calculations.

† For the treatment of the Coulomb field, see [Qu 72, GVL 73, Sl 51, Go 52, KS 72, TQ 74].

(iii) the so-called spin orbit densities

$$\mathbf{J}(\mathbf{r}) = (-i) \sum_{itss'} \varphi_i^*(\mathbf{r}, s, t)\left[\nabla \varphi_i(\mathbf{r}, s', t) \times \sigma_{ss'}\right]. \tag{5.86}$$

The sums are taken over all occupied single-particle states. A lengthy but straightforward calculation [VB 72] for $N = Z$ nuclei gives

$$H(\mathbf{r}) = \frac{\hbar^2}{2m}\tau(\mathbf{r}) + \frac{3}{8}t_0\rho^2 + \frac{1}{16}t_3\rho^3 + \frac{1}{16}(3t_1 + 5t_2)\rho\tau + \frac{1}{64}(9t_1 - 5t_2)(\nabla\rho)^2 - \frac{3}{4}W_0\rho\nabla\mathbf{J} + \frac{1}{32}(t_1 - t_2)\mathbf{J}^2. \tag{5.87}$$

Besides the kinetic energy τ, we also have contributions from the two-body δ-force $\sim\rho^2$ and the three-body δ-force $\sim\rho^3$. The nonlocal p^2-terms give contributions $\sim\rho\tau$ and $\sim\nabla\rho^2$. The latter has its largest contributions at the nuclear surface. The term $\frac{1}{32}(t_1 - t_2)\mathbf{J}^2$ is usually neglected because it is difficult to handle in deformed nuclei and its contribution to the spin orbit part does not reproduce the experimental spin-orbit splitting.

We could also have derived the three-body term $\sim\rho^3$ from a density dependent two-body interaction

$$\frac{1}{16}t_3\rho^3 = \frac{1}{2}\sum_{i,j<A}\langle ij|\frac{1}{6}t_3\delta(\mathbf{r}_1 - \mathbf{r}_2)\rho(r_1)(1 + P^\sigma)|ij - ji\rangle. \tag{5.88}$$

In Section (5.6.1.1) we saw that when using density dependent interactions we have first to calculate the energy and only afterwards vary with respect to the density. In that sense, the three-body contact force of Skyrme is equivalent to the two-body interaction (5.88). This equivalence, however, is only valid in even–even nuclei with time reversal symmetry.

Using (5.87), we are able to calculate the binding energy per particle in nuclear matter without Coulomb interaction. In this case we have translational invariance and the single-particle wave functions are given by plane waves normalized to a δ-function

$$\varphi_{kst} = \frac{1}{(2\pi)^{3/2}} e^{i\mathbf{kr}} \chi_s^{1/2} \chi_t^{1/2} \tag{5.89}$$

and in Eqs. (5.84)–(5.86) we have to replace

$$\sum_{i=1}^{A} \ldots \quad \text{by} \quad \int_{|k|<k_F} d^3k \ldots, \tag{5.90}$$

where all the levels with $|k|$ smaller than the Fermi momentum k_F are occupied. From (5.84), we get the usual relation between ρ and k_F [see also Eqs. (13.22) and (13.23)]

$$\rho = \frac{4}{(2\pi)^3}\frac{4\pi}{3}k_F^3 = \frac{2}{3\pi^2}k_F^3 \tag{5.91}$$

and from (5.85),

$$\tau = \frac{2}{3\pi^2}\frac{3}{5}k_F^5 = \frac{3}{5}\rho k_F^2 = \frac{3}{5}\left(\frac{3\pi^2}{2}\right)^{2/3}\rho^{5/3}. \tag{5.92}$$

Because of translational invariance, we have $\nabla\rho = \nabla\mathbf{J} = 0$ and obtain for the binding energy per particle in nuclear matter

$$\frac{E_0}{A} = \frac{H}{\rho} = \frac{3}{5}\frac{\hbar^2}{2m}k_F^2 + \frac{3}{8}t_0\rho + \frac{1}{16}t_3\rho^2 + \frac{3}{80}(3t_1+5t_2)\rho k_F^2. \quad (5.93)$$

The saturation property means that there is an equilibrium density ρ_0 for which

$$\frac{\partial}{\partial\rho}\left(\frac{E_0}{A}\right)\Bigg|_{\rho=\rho_0} = 0 = \frac{2}{5}\frac{\hbar^2}{2m}k_F^2\rho^{-1} + \frac{3}{8}t_0 + \frac{1}{8}t_3\rho + \frac{1}{16}(3t_1+5t_2)k_F^2. \quad (5.94)$$

The incompressibility of nuclear matter K is defined as the curvature of the binding energy E_0/A with respect to the Fermi momentum k_F at this minimum:

$$K = k_F^2\frac{\partial^2(E_0/A)}{\partial k_F^2}\Bigg|_{\rho=\rho_0} = \frac{6}{5}\frac{\hbar^2}{2m}k_F^2 + \frac{9}{4}t_0\rho + \frac{15}{8}t_3\rho^2 + \frac{3}{4}(3t_1+5t_2)\rho k_F^2. \quad (5.95)$$

Equations (5.93)–(5.95) allow us to express the two constants t_0 and t_3 and the combination $3t_1+5t_2$ by the nuclear matter constants E_0/A, ρ_0, and K. From Eq. (5.95), we see that t_3 is strongly correlated with the incompressibility K.* From the Bethe–Weizsäcker Formula (1.4), we know that the value of $E_0/A = a_V = 15.9$ MeV. Less well determined is the equilibrium density $\rho_0 = 3/4\pi r_0^3 \simeq 0.14$ fm^{-3}. Therefore, it is not possible to adjust the force parameters of a phenomenological force to nuclear matter data alone. We first have to carry out the calculation for finite nuclei.

5.6.2.3. The Derivation of the Density Dependent Hartree–Fock Equations. According to the concept of Section 5.6.1, we have to vary the functional $E_0[\rho]$ with respect to the density in order to gain the Hartree–Fock Hamiltonian. Unfortunately, (5.95) does not have the form of a functional of ρ. It also depends on τ and $\mathbf{J}$, and it is very hard to express τ and $\mathbf{J}$ in terms of ρ. In our case, however, this is no real problem, since the HF-density is uniquely defined by the single-particle wave functions φ_k (5.84), and we can also carry out the variation with respect to φ_k under the condition that they are normalized to unity. We use Lagrange multipliers ϵ_k for these subsidiary conditions and find

$$\frac{\delta}{\delta\varphi_k}\left(E_0[\rho] - \sum_l \epsilon_l \int d^3r|\varphi_l(\mathbf{r})|^2\right) = 0. \quad (5.96)$$

The variation of the energy (5.83), after integrating by parts, can be written

$$\delta E = \int d^3r\left[\frac{\hbar^2}{2m^*(\mathbf{r})}\delta\tau(\mathbf{r}) + U(\mathbf{r})\delta\rho(\mathbf{r}) + \mathbf{W}(\mathbf{r})\delta\mathbf{J}(\mathbf{r})\right] \quad (5.97)$$

*Nuclear matter calculations give values between 150 and 250 MeV and, from recent measurements of the breathing mode in ^{208}Pb, we deduce the value $K \simeq 200$ MeV for the incompressibility of heavy, finite nuclei.

with an effective mass

$$m^*(\mathbf{r}) = m\left(1 + \frac{2m}{\hbar^2}\frac{1}{16}(3t_1 + 5t_2)\rho\right)^{-1}, \tag{5.98}$$

an average field

$$U(\mathbf{r}) = \frac{3}{4}t_0\rho + \frac{3}{16}t_3\rho^2 + \frac{1}{16}(3t_1 + 5t_2)\tau + \frac{1}{32}(5t_2 - 9t_1)\nabla^2\rho - \frac{3}{4}W_0\nabla\mathbf{J}, \tag{5.99}$$

and a one-body spin-orbit potential [we neglect the term $\frac{1}{32}(t_1 - t_2)\mathbf{J}^2$]

$$\mathbf{W}(\mathbf{r}) = \frac{3}{4}W_0\nabla\rho. \tag{5.100}$$

We now have to insert into Eq. (5.96) the variations $\delta\tau$, $\delta\rho$, and $\delta\mathbf{J}$ with respect to φ_k. From definitions (5.84)–(5.86) we get

$$\delta E = 2\sum_{i=1}^{A}\int d^3r\,\delta\varphi_i^*\left\{-\nabla\frac{\hbar^2}{2m_r^*}\nabla + U + \mathbf{W}\frac{1}{i}(\nabla\times\boldsigma)\right\}\varphi_i \tag{5.101}$$

and, using Eq. (5.96), we finally find the HF-equation in coordinate space viz:

$$\left\{-\nabla\frac{\hbar^2}{2m^*(\mathbf{r})}\nabla + U(\mathbf{r}) + \mathbf{W}\frac{1}{i}(\nabla\times\boldsigma)\right\}\varphi_i(\mathbf{r}) = \epsilon_i\varphi_i(\mathbf{r}). \tag{5.102}$$

The exchange term in the HF-equation (5.44) is now local and is included in the potential $U(\mathbf{r})$, so that (5.102) is a pure differential equation. The nonlocality is expressed only in the $\mathbf{r}$-dependence of the effective mass $m^*(\mathbf{r})$. In the case of spherical symmetry, we end up with a one-dimensional differential equation of second order in the radial coordinate r. In particular, the spin-orbit term (5.100) takes the form

$$\frac{3}{2}W_0\left(\frac{1}{r}\frac{\partial}{\partial r}\rho\right)\mathbf{ls} \tag{5.103}$$

and, as we have already discussed in Section 2.4, this is concentrated mainly at the nuclear surface.

5.6.2.4. Discussion of the Results. Vautherin and Brink [VB 72] originally solved the HF equation (5.102) for the spherical closed shell nuclei ^{16}O, ^{40}Ca, ^{48}Ca, ^{90}Zr, and ^{208}Pb, and were able to adjust the six parameters t_0, t_1, t_2, t_3, W_0, and x_0 so as to reproduce the radii and binding energies of these nuclei. They presented two sets of force constants (Skyrme I and II) which gave a good description of these closed shell nuclei. Both have large values of t_3, which means a strong density dependence. This is a very crucial point, because for density dependent forces the binding energy E_0^{HF}

is given by (5.28) and (5.81)

$$E_0^{\mathrm{HF}} = \mathrm{Tr}\, t\rho + \frac{1}{2}\mathrm{Tr}_1\mathrm{Tr}_1 \rho\bar{v}\rho$$

$$= \frac{1}{2}\mathrm{Tr}\{(t+\epsilon)\rho\} - \frac{1}{2}\mathrm{Tr}\left(\langle \mathrm{HF}|\frac{\delta V}{\delta \rho}|\mathrm{HF}\rangle \rho\right) \tag{5.104}$$

which, in the case of the Skyrme force (5.88), is

$$E_0^{\mathrm{HF}} = \frac{1}{2}\sum_{i=1}^{A}(t_{ii}+\epsilon_i) - \frac{1}{32}t_3\int d^3r\, \rho^3. \tag{5.105}$$

The last term in Eqs. (5.104) and (5.105) is, as discussed in Section 5.6.1.2, called the rearrangement term. For a density independent force it vanishes, and it is not possible to simultaneously reproduce the binding energy E_0, the radii, and the experimental single-particle energies ϵ_i of a nucleus, as the last two quantities already determine (more or less) the first part of Eq. (5.105). Only the rearrangement term makes it possible to reproduce the binding energy so well. This term is always negative because of the replusion of the nuclear forces at short distances or large densities.

Later, the density dependence in many other spherical nuclei was investigated [BFG 75] and other sets of force parameters were determined (Skyrme III–VI). They differ in their density dependence, and it turns out that t_3 is not determined by the radii and binding energies alone. With rather different values of t_3, we can, in fact, reproduce these values. However, the single-particle energies ϵ_i do depend dramatically on t_3. Skyrme III [Eq. (4.108)] gives reasonable values for all these quantities.

Table 5.3 lists binding energies and root mean square radii r_c for several spherical closed-shell nuclei as calculated by the most sophisticated

Table 5.3 Experimental and calculated root mean square radii (in fm) and binding energies (in MeV) per nucleon

		Experiment	Negele [Ne 70]	Campi and Sprung [CS 72]	Nemeth et al. [NMH 73]	Skyrme III [BFG 75]
^{16}O	E	−7.98	−6.75	−7.68	−7.98	−7.96
	r_c	2.73	2.80	2.75	2.77	2.69
^{40}Ca	E	−8.55	−7.49	−8.33	−8.47	−8.54
	r_c	3.49	3.49	3.49	3.40	3.48
^{48}Ca	E	−8.67	−7.48	−8.40	−8.55	−8.71
	r_c	3.48	3.52	3.51	3.44	3.53
^{90}Zr	E	−8.71	−7.85	−8.63	−8.70	−8.71
	r_c	4.23	4.25	4.27	4.13	4.32
^{208}Pb	E	−7.87	−7.53	−7.87	−7.87	−7.87
	r_c	5.50	5.44	5.45	5.22	5.57

Hartree–Fock calculations with effective interactions derived from the bare nucleon–nucleon force and also with Skyrme III.

One has also calculated angular distributions for electron scattering and found good agreement with the experimental data. This shows that the calculations produce the proper charge distributions. For example, in Fig. 5.5(a) we show the charge distribution of ^{208}Pb compared with a phenomenological curve determined from electron scattering. Figure 5.5(b) gives the corresponding average potential U_p and U_n for protons and neutrons and the effective mass m^*/m. The HF-charge density is not completely constant in the nuclear interior, but shows some oscillations which have their origin in the shell effects. They are smaller than the oscillations that would be obtained for a shell model charge density in a phenomenological Wood–Saxon potential, but are still larger than the experimentally observed wiggles. The corresponding self-consistent fields also show deviations from a Wood–Saxon shape. For lighter nuclei, they are more important. Figure 5.5(b) also gives the ratio m^*/m, which measures the nonlocality of the Skyrme potential. It is given by the parameter combination $3t_1+5t_2$ [see Eq. (5.98)]. For Skyrme II, the nonlocality is well pronounced.

The effective mass has a strong influence on the single particle energies ϵ_i. If we assume the single-particle wave functions of the A and the $A-1$ system to be identical, that is, if we neglect the polarization of the core by the hole in the level i, then we get for the energy difference

$$E(A)-E_i(A-1)=t_{ii}+\sum_{j<A}\bar{v}^{(2)}_{ij,ij}+\frac{1}{2}\sum_{jk<A}\bar{v}^{(3)}_{ijk,ijk}. \tag{5.106}$$

The evaluation of the right-hand side shows [VB 72] that (5.106) is exactly the single-particle energy ϵ_i, the eigenvalue of the HF-Hamiltonian. In Fig. 5.6, therefore, we compare the experimental single-particle levels in ^{208}Pb with the calculated single-particle energies ϵ_i. Essentially, we obtain the

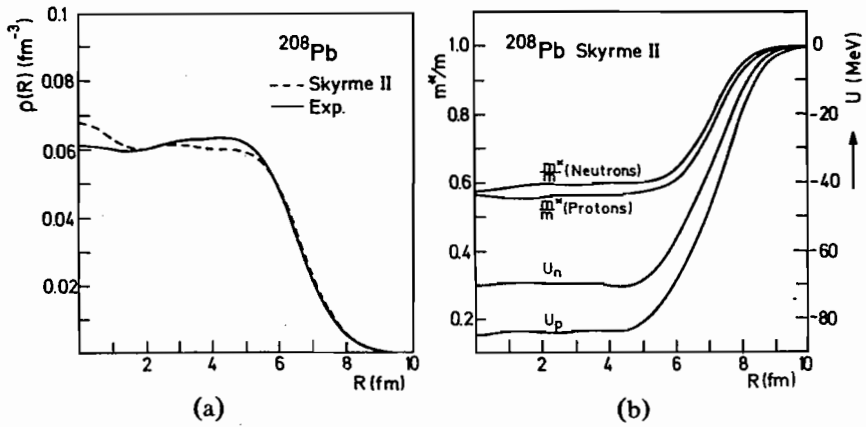


Figure 5.5. Hartree–Fock results for ^{208}Pb with the interaction Skyrme II. (a) Charge distribution. (b) Effective mass m^*/m and HF-potential $U(r)$ (the proton single-particle potential does not include the Coulomb term). (From [VB 72].)

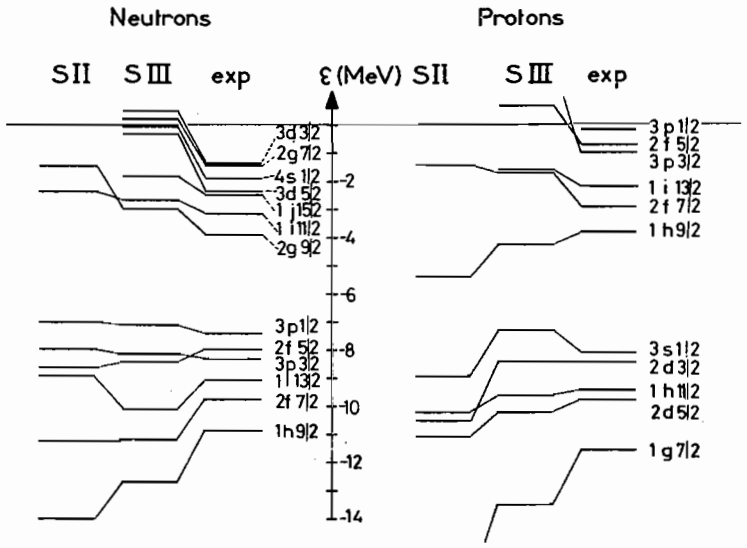


Figure 5.6. Experimental and calculated single-particle energies in the lead region. The calculated values were obtained with two different versions of the Skyrme force. With increasing effective mass, we recognize a compression of the spectrum ($m^*/m \simeq 0.6$ for SII and $\simeq 0.75$ for SIII). (From [BFG 75].)

correct ordering of the single-particle levels, although quantitatively the different sets of parameters show rather different spectra. However, it should be emphasized that we cannot expect a priori complete agreement, because polarization effects play a certain role that have not yet been taken into account [RW 73].

5.7 Concluding Remarks

Summarizing the results of this chapter, we can say that there is a microscopic foundation of the spherical shell model as discussed in Chapter 2. The Hartree–Fock method provides a tool for calculating the average potential from an effective nucleon interaction. It becomes very important to use density dependent effective interactions to get quantitative agreement. In this case, we are able to reproduce the ground state properties of spherical nuclei very well. In most cases, the calculated single-particle levels show the correct ordering; however, they deviate in detail from the experimental single-particle excitations of neighboring odd mass nuclei.

So far we discussed only applications of the HF method to nuclei with closed shells. In fact, these are the only cases where the self-consistent field can have spherical symmetry. As soon as one or several particles are put in an unfilled j-shell (*open shell* HF [Ke 63]) we have to decide which one of the orginally degenerate magnetic quantum numbers m needs to be occu-

pied and we get a slightly deformed density distribution (oblate with respect to the quantization axis for large values of $|m|$ and prolate for small values of $|m|$). This deformed density produces a deformed mean field in the next step of the iteration, which also changes the single-particle wave functions in the core. In this way, we take into account the correlations caused by the interaction of the external particle with the other nucleons. We call this effect a *polarization* of the core.

As long as we have only one or a few external particles, we can also treat this interaction in the spherical basis by a shell model configuration mixing calculation allowing for *ph excitations*. In this picture, the valence nucleons excite virtual vibrations in the core and we treat the polarization effects by a particle vibrational coupling technique (see Sec. 9.3).

In many practical applications, this effect has been taken into account only in an averaged way by distributing the external particle over all the m-quantum numbers of the next higher j-shell with equal probability $v_m^2 = 1/(2j+1)$. The wave function is then, in a sense, a HFB state (see Chap. 7) with a spherical density distribution.

In principle, the mean field approach is only justified if there is a well pronounced energy gap between the highest occupied level and the first empty level. If that is not the case (subshell closures), it is easy to excite virtual *ph*-pairs, and we can expect a more complicated wave function than a Slater determinant [see Eq. (2.36)]. From these arguments, we expect that the HF method yields a better approximation to the exact ground state for magic nuclei than for nuclei with a few particles away from the closed shell configuration. These nuclei show a small deformation in HF and have many nearly degenerate levels in the vicinity of the Fermi surface.

We call such nuclei transitional nuclei, and in the Chapters 9 and 10 we will discuss some methods for investigating their structure. For nuclei far from closed magic configurations, however, the correlation among the quasi-particles becomes so strong that they can again be treated in an extended mean field approach. Depending on the kind of correlations, we have to use a *deformed* HF *potential*—which is again a very good approximation for cases where new magic numbers develop in the deformed region (see Fig. 2.25). We will discuss these methods together with the nuclear pairing phenomenon in more detail in Chapters 6 and 7, and we will see then that nuclear deformations with density dependent forces can be explained rather nicely.